Assignment No	**8**
Title	**Linked list**
Objective	**Singly Linked list- Insertion**
Roll No	**MCA2649**

Source Code:-

```
//Name:- Satish Rathod
//Roll No:- 2649
//LinkedList


//1)Singly Linked list
//Design a Node class
class Node
{
	int data;
	Node next;
	Node(int data)
	{
		this.data=data;
		this.next=null;
	}
}
class SinglyLinkedList
```

```
{
    // define head
    private Node head=null;
    // a method to check is list empty
    public boolean isEmpty()
    {
        return head==null;
    }
    // insert at the begining
    public void insertFirst(int data)
    {
        //creat node
        Node temp = new Node(data);
        //link to old head
        temp.next = head;
        head = temp; // move the head
    }
    //insert at the end
    public void insertLast(int data)
    {
        //create node
        Node temp = new Node(data);
        //check is the list empty
        if(isEmpty())
        {
```

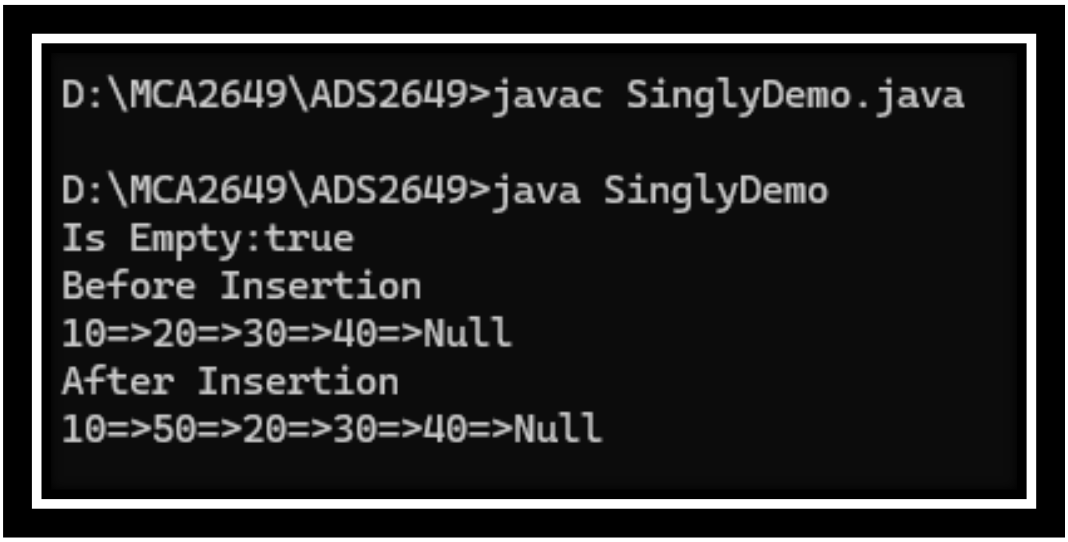
```
                head=temp;
        }
        // traverse the list till end
        Node tr = head; //start traversal
        while(tr.next != null)
        {
                tr = tr.next; // walk till end and stop at end
        }
        tr.next=temp; // assign new node at the end
}
// Display the list
public void display()
{
        Node temp=head;
        while(temp != null)
        {
                //print each node
                System.out.print(temp.data + "=>");
                // move to the next node
                temp = temp.next;
        }
        System.out.println("Null");

}
//Insert at given position
```

```
public void insertAt(int pos,int data)
{
        if(pos == 1)
        {
                //call insert First
                insertFirst(data);
                return;
        }
        // traverse to the specified position
        Node temp = head;
        for(int i=1; i<(pos-1) && (temp!=null); i++)
        {
                temp=temp.next;
                if(temp==null)
                {
                        System.out.println("Invalid position!");
                        return;
                }
        }
        // assign new node
        Node newN = new Node(data);
        newN.next=temp.next; // assign the newnode to old
        temp.next=newN; // assign new node at specified position
}
```

```
}
class SinglyDemo
{
	public static void main(String args[])
	{
		SinglyLinkedList l1 = new SinglyLinkedList();
		System.out.println("Is Empty:"+l1.isEmpty());
		l1.insertFirst(10);
		l1.insertLast(20);
		l1.insertLast(30);
		l1.insertLast(40);
		System.out.println("Before Insertion");
		l1.display();
		System.out.println("After Insertion");
		l1.insertAt(2,50);
		l1.display();
	}
}
```

OUTPUT:-


D:\MCA2649\ADS2649>javac SinglyDemo.java
D:\MCA2649\ADS2649>java SinglyDemo
Is Empty:true
Before Insertion
10=>20=>30=>40=>Null
After Insertion
10=>50=>20=>30=>40=>Null